RANJANI KOUSHIK

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Education

Georgia Institute of Technology Aug 2023 - May 2027
Bachelor of Science in Computer Science, GPA 3.75 Atlanta, GA

Experience

Google Research, Student Researcher
May 2026 – Present
MoQ, WebRTC, Linux tc/NetEm, C++, Cloud Seattle, WA

- Researched network protocols to support a seven-camera 4K arrangement for 3D holographic video at Google Beam.
- Prototyped Media over QUIC and benchmarked it against WebRTC using network disruption experiments.
- Reduced frame drops by 20% and latency by 17% by designing bandwidth estimation algorithms.

DaVita, Software Engineering Intern
May 2025 – Aug 2025
Python, SQL, AI Denver, CO

- Automated dialysis facility placement from hospital referrals by architecting a Gemini-powered multi-agent workflow with LangGraph, Google ADK, FastAPI, and Streamlit.
- Agents configured to communicate over chat, email and phone call (Twillio), reducing average patient placement time from 4 days to 12 minutes across across 200 test referrals.

Siemens, Software Engineer Intern
May 2024 – Aug 2024
C++, React, Node.js, GitLab CI/CD Boulder, CO

- Automated vulnerability detection for electronic design automation software, saving analysts 60 hours per month.
- Did this by building a web app that scanned code with Black Duck and pulled vulnerability details from NIST NVD.

Georgia Tech Research Institute, Research Intern
Aug 2025 – Jan 2026
ResNet, Mask R-CNN, SAM2 Atlanta, GA

- Researched early detection of tomato spotted wilt virus, a disease causing significant peanut yield and economic loss.
- Built ML pipelines cutting detection time from 4 weeks to 1 week, and improving model accuracy from 54% to 86%.

Healiam, Medical Information Tech Startup Founder
Aug 2022 – Aug 2025
Pyserial Boulder, CO

- Founded a blood testing startup that reached $100,000 in annual recurring revenue.
- Named to Colorado Inno's Top 25 Under 25 Entrepreneurs

Research

Georgia Tech, Murty Lab, Cognition and Computation Research Intern
Feb 2024 – Present
PyTorch, scikit-learn Atlanta, GA

- Built a brain-response prediction model that mapped visual scenes to fMRI activity, achieving an average correlation of 0.87 across brain regions.
- Developed research pipelines combining encoding models, clustering, and high-throughput screening to identify a previously uncharacterized subdivision of the parahippocampal place area.

Georgia Tech, PAIR Lab, Robotics Research Intern
Apr 2024 – Jun 2025
Python, Flutter, Isaac Sim, TCP/WebRTC Atlanta, GA

- Accelerated robot training data collection by 10× over physical collection by building a platform for controlling simulated robots using VR, keyboard, and SpaceMouse.
- Built a TCP/WebRTC pipeline to stream robot commands and synchronize live visual feedback.

"COBALT: Crowdsourcing Robot Learning via Cloud-Based Teleoperation with Smartphones"
IEEE International Conference on Robotics & Automation 2026

Also loves: operating systems, watercolouring, racquetball, and thrifting!